

SAI PRASHANTH NAGISETTI

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PROFESSIONAL SUMMARY

Generative AI Engineer with 4+ years of experience in AI-driven solutions through machine learning models and application development. Proficient in Conversational AI, Retrieval-Augmented Generation (RAG), and LLM integration. Skilled with modern AI stacks including LangChain, LangGraph, Vector DBs, and Knowledge Graphs. Adept at integrating AI systems into robust enterprise ecosystems with a focus on security, performance and scalability. Skilled in implementing and optimizing REST APIs for performance, security, and scalability.

Technical Skills

- **Programming Languages:** Python, Java, SQL, Unix Shell.
- **Machine Learning & Deep Learning:** Supervised & Unsupervised Learning, Predictive Modeling, Feature Engineering, Hypothesis Testing, Neural Networks, CNN, RNN, LSTM, NLP, MLflow, Transfer Learning, Inferential Statistics.
- **Generative AI:** Transformers, LLM's, BERT, GPT, DALL-E, LangChain, RAG, LlamaIndex, Vector Databases, AI Agents, Tools, AutoGen, CrewAI, LangGraph, Hugging Face, OpenAI API, Prompt Engineering, RAGAS.
- **Libraries:** Pandas, NumPy, Matplotlib, Scikit-Learn, SpaCy, TensorFlow, Keras, PyTorch, OpenCV.
- **Data Visualization:** Statistical Analysis, Power BI, Grafana, Excel, Tableau.
- **Software Development & Web Technologies:** REST APIs, Flask, Streamlit, HTML, CSS, JavaScript, Next.js.
- **Cloud Platforms:** Azure, AWS (Amazon Bedrock, Sagemaker, Lambda, EC2, S3, CloudWatch, API Gateway, Amazon Lex).
- **Tools & Technologies:** MySQL, PL/SQL, MongoDB, Neo4j, GitHub, Kubernetes, Docker, Jenkins, Elasticsearch.

WORK EXPERIENCE

Client - United Health Care/ Minnetonka, MN

Jul 2024 – Present

Role: Generative AI Engineer

- Developed conversation logic for an AI-powered chatbot using **AWS Lex** and **GPT-4 model**, enabling end-to-end customer support for policy management, claim tracking, and customer queries.
- Integrated **Retrieval-Augmented Generation (RAG)** to dynamically fetch policy documents, claim guidelines, and FAQs, improving chatbot accuracy and reducing agent escalations.
- Designed and implemented **REST API** integrations using **AWS API Gateway** and **Lambda**, enabling real-time claim status checks and policy updates.
- Managed scalable data storage using **AWS DynamoDB**, ensuring structured and secure data handling for interactions.
- Monitored system performance with **AWS CloudWatch**, ensuring proactive error detection and optimizing response times.

Client - British Telecom Services, UK /TCS, Hyderabad, India

Nov 2020 – Oct 2022

Role: Data Analyst

- Developed and optimized predictive models (**Logistic Regression, XGBoost, Random Forest**) on 300K+ customer records, achieving an **F1-score of 0.82** and improving churn prediction accuracy by 10%.
- Automated end-to-end model deployment using Docker, **Airflow**, and Jenkins, reducing deployment time by 40% and ensuring real-time inference scalability.
- Built interactive **Power BI, Grafana** dashboards to monitor churn trends and model performance, reducing decision-making time by 35%.

Client - Verizon Communications, USA / TCS, Hyderabad, India

Sep 2019– Nov 2020

Role: Data Analyst

- Implemented **K-Means clustering** to segment customers based on service usage, spending patterns and demographics, enhancing targeted marketing and customer engagement.
- Performed data preprocessing and **Exploratory Data Analysis (EDA)** using **Pandas, Seaborn, and Matplotlib**, identifying key trends that led to a **15%** improvement in segmentation accuracy.
- Developed predictive models (**Random Forest, Logistic Regression**) to forecast churn likelihood and upsell potential, reducing **Mean Absolute Error** by **20%**, increasing campaign responses by **12%**, and improving customer retention.

Role: Software Engineer

- Designed and deployed a high-performance **REST API** using **Python** and **Flask**, reducing response latency by 40%, following **Software Development Life Cycle (SDLC)** principles.
- Optimized **SQLite database** indexing, accelerating data retrieval speeds by 35%.
- Strengthening application security by implementing **session-based authentication**, preventing unauthorized access.
- Achieved **90% test coverage** through rigorous unit tests with **unittest**, ensuring API reliability and performance.
- Documented API endpoints** and usage guidelines, enhancing developer accessibility and collaboration.

PROJECTS

AI-Powered Content Generation System for Tech Blogs | Technologies Used: Python, AI, CrewAI, GPT-4
Developed an automated content generation system that integrates AI agents for blog research and writing. The project leverages a YouTube search tool to gather video transcriptions on tech topics and uses AI-powered agents to generate detailed blog posts.

RAG-based Document Retrieval System using LangGraph and AstraDB
Built a Retrieval-Augmented Generation (RAG) pipeline using LangGraph and LangChain to fetch context-aware answers from technical blogs. Used HuggingFace embeddings and Chroma for text chunking, and integrated AstraDB as the vector store for scalable semantic search. Implemented document loading, embedding generation, and natural language querying for real-time information retrieval.

Evaluating the Efficacy of Different Language Models on Dementia Prediction
Analyzed the performance of six NLP models (**DAN, BERT, RoBERTa, Longformer, Clinical Longformer, and Big-BIRD**) for dementia prediction from transcribed patient conversations. Fine-tuned and evaluated models using **precision, recall, F1-score** and **accuracy** on a balanced dataset (53% dementia, 47% control). Achieved the highest accuracy (88%) with RoBERTa and BigBIRD, demonstrating the potential of optimized transformers for early detection of cognitive decline.

Generative AI-Powered Market Analysis Tool
Developed a Generative AI-powered market analysis tool leveraging **AWS Bedrock, Amazon Knowledge Bases**, and **LangChain** for real-time insights and predictive analytics. Implemented interactive dashboards using Streamlit for custom visualizations and financial data analysis. Integrated AI-driven document summarization and Q&A capabilities using AWS Bedrock Runtime APIs and Boto3.

Fuel Price Prediction
Forecasted fuel prices using historical data and models such as **Linear Regression, Decision Tree, Random Forest, and XGBoost**. Conducted EDA to identify correlations, such as the impact of crude oil prices on fuel costs. Optimized model parameters using **GridSearchCV**, with **XGBoost** achieving the highest performance ($R^2 = 0.9875$).

EDUCATION

University of Missouri-Kansas City	Aug 2022 – May 2024
Master of Computer Science (Specialization in Data Science)	
K L University	June 2015 – May 2019
Bachelor of Technology in Electronics and Communication Engineering	

CERTIFICATIONS

- AI Agentic Design Patterns with AutoGen – DeepLearning.ai
- Develop applications with Azure OpenAI Service
- AWS Certified Machine Learning - Specialty